

Software Quality - W8.A6

Model Based Testing

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1 Part I: Decision Table for Car Rental Pricing

The decision table below tests the **Car Rental Pricing Rules**. The first column lists the elementary conditions, while each remaining column is a test-case specification. $\$x$ represents the gross rental cost (*base daily rate* $\times$ *number of days*), $\$p$ the discounted price obtained after applying the duration and customer-type discounts and n is the number of rental days.

For readability the table is presented in two blocks: the first block lists columns C1 to C9, the second block continues with columns C10 to C15. The two blocks form a single decision table; the row labels in the first column are the same in both blocks. A dash represents a don't care entry.

Condition	C1	C2	C3	C4	C5	C6	C7	C8	C9
Economy	-	-	F	-	-	F	-	-	F
Standard	-	-	T	-	-	-	-	-	-
SUV	F	F	F	-	-	-	-	-	-
1 to 2 days	T	T	T	F	F	F	F	F	F
3 to 5 days	F	F	F	T	T	T	F	F	F
6+ days	F	F	F	F	F	F	T	T	T
Regular	T	F	F	T	F	F	T	F	F
Member	F	T	F	F	T	F	F	T	F
VIP	F	F	T	F	F	T	F	F	T
Weekday pickup	T	T	T	T	T	T	T	T	T
Weekend pickup	F	F	F	F	F	F	F	F	F
Airport pickup	F	F	F	F	F	F	F	F	F
GPS	F	F	F	F	F	F	F	F	F
Child Seat	F	F	F	F	F	F	F	F	F
Additional Driver	F	F	F	F	F	F	F	F	F
Early Return	F	F	F	F	F	F	F	F	F
Total price	$\$x$	$\$x-5\%$	$\$x-12\%$	$\$x-10\%$	$\$x-14.5\%$	$\$x-20.8\%$	$\$x-20\%$	$\$x-24\%$	$\$x-29.6\%$

Condition	C10	C11	C12	C13	C14	C15
Economy	-	F	-	-	-	-
Standard	-	-	-	-	-	-
SUV	-	-	-	-	-	-
1 to 2 days	-	-	-	-	-	-
3 to 5 days	-	-	-	-	-	-
6+ days	-	-	-	-	-	-
Regular	-	-	-	-	-	-
Member	-	-	-	-	-	-
VIP	-	-	-	-	-	-
Weekday pickup	F	F	-	-	-	-
Weekend pickup	T	F	-	-	-	-
Airport pickup	F	T	-	-	-	-
GPS	F	F	T	F	F	F
Child Seat	F	F	F	T	F	F
Additional Driver	F	F	F	F	T	F
Early Return	F	F	F	F	F	T
Total price	$1.10 \$p$	$\$p+25$	$\$p+8n$	$\$p+5n$	$\$p+15n$	$\$p-\ref

Legend

- $\$x$: gross rental cost, equal to *base daily rate* $\times$ *number of days*, where Economy is \$40/day, Standard is \$60/day and SUV is \$90/day.
- $\$p$: price after the duration discount and the customer-type discount have been applied.
- n : number of rental days. Add-ons such as GPS \$8/day, Child Seat \$5/day, Additional Driver \$15/day, are charged for every rental day and are never discounted.
- k : number of *unused* prepaid days returned by the customer ($0 < k < n$).
- $\$ref$: refund of the unused prepaid days at the discounted daily rate. $\$ref = \frac{\$p}{n} \cdot k$. No early-return penalty is charged.
- Combined discount values: at 1 to 2 days Member = 5%, VIP = 12%; at 3 to 5 days Regular = 10%, Member = $1 - 0.9 \cdot 0.95 = 14.5\%$, VIP = $1 - 0.9 \cdot 0.88 = 20.8\%$; at 6+ days Regular = 20%, Member = $1 - 0.8 \cdot 0.95 = 24\%$, VIP = $1 - 0.8 \cdot 0.88 = 29.6\%$.

1.1 Constraints

- exactly-one (Economy, Standard, SUV)
- exactly-one (1 to 2 days, 3 to 5 days, 6+ days)
- exactly-one (Regular, Member, VIP)
- exactly-one (Weekday pickup, Weekend pickup, Airport pickup)
- SUV - (3 to 5 days $\vee$ 6+ days) SUV requires a minimum of 3 days
- VIP - (Standard $\vee$ SUV) VIP cannot rent Economy
- Airport pickup - (Standard $\vee$ SUV) Airport pickup is not available for Economy

1.2 Optional: test cases for single-condition coverage

Single-condition coverage is satisfied by 15 test cases that correspond one-to-one to the 15 columns of the decision table.

TC	Vehicle	Days	Customer	Pickup	Add-ons / Early ret.	Expected price
T1	Economy	2	Regular	Weekday	none	\$80
T2	Standard	2	Member	Weekday	none	$\$120 - 5\% = \114
T3	Standard	2	VIP	Weekday	none	$\$120 - 12\% = \105.60
T4	Economy	4	Regular	Weekday	none	$\$160 - 10\% = \144
T5	Standard	4	Member	Weekday	none	$\$240 - 14.5\% = \205.20
T6	SUV	4	VIP	Weekday	none	$\$360 - 20.8\% = \285.12
T7	Economy	7	Regular	Weekday	none	$\$280 - 20\% = \224
T8	Standard	7	Member	Weekday	none	$\$420 - 24\% = \319.20
T9	SUV	7	VIP	Weekday	none	$\$630 - 29.6\% = \443.52
T10	Standard	3	Regular	Weekend	none	$\$162 \cdot 1.10 = \178.20
T11	Standard	3	Regular	Airport	none	$\$162 + \$25 = \$187$
T12	Standard	3	Regular	Weekday	GPS	$\$162 + 8 \cdot 3 = \186
T13	Standard	3	Regular	Weekday	Child Seat	$\$162 + 5 \cdot 3 = \177
T14	Standard	3	Regular	Weekday	Additional Driver	$\$162 + 15 \cdot 3 = \207
T15	Standard	6	Regular	Weekday	Early Return ($k=2$)	$\$288 - 96 = \192

2 Part II: Grammar for HR Employee Profile

The grammar derives strings that conform to the **HR Employee Profile** XSD schema. Bounded recursive productions are annotated in square brackets [k] with the *maximum number of times* the production may be applied. For collections that the schema declares `maxOccurs="unbounded"`, such as **skills** and **projects**, a conventional bound [10] is adopted for boundary-condition test generation, in line with the W7.H6 sample solution.

```

<employee>                ::=    <emp_id> <first_name> <last_name> <email>
                                <departments> <skills> <certifications> <projects>

<emp_id>                   ::=    integer
<first_name>               ::=    string
<last_name>                ::=    string
<email>                    ::=    string

<departments>              ::=    <departmentList>
<departmentList>           ::=    <department>
[3] <departmentList>       ::=    <department> <departmentList>
<department>               ::=    string

<skills>                   ::=    <skillList>
<skillList>                ::=    <skillInfo>
[10] <skillList>           ::=    <skillInfo> <skillList>
<skillInfo>                ::=    <skill_name> <proficiency> <years_experience>
<skill_name>               ::=    string
<proficiency>              ::=    string
<years_experience>          ::=    integer

<certifications>           ::=    <certList>
<certList>                 ::=    <certInfo>
[5] <certList>             ::=    <certInfo> <certList>
<certInfo>                 ::=    <cert_id> <cert_name> <issue_date> <expiry_date>
<cert_id>                  ::=    integer
<cert_name>                ::=    string
<issue_date>               ::=    date
<expiry_date>              ::=    date

<projects>                 ::=    <projectList>
<projectList>              ::=    <projectInfo>
[10] <projectList>         ::=    <projectInfo> <projectList>
<projectInfo>              ::=    <project_id> <project_name> <role> <start_date>

```

<project_id>	::=	integer
<project_name>	::=	string
<role>	::=	string
<start_date>	::=	date